

SCUT Data Analysis And Modeling

Lab 5: Foundations of Statistical Inference - Confidence Intervals

Use the template

Before you begin the lab, don't forget to use the template used in the previous lab lesson.

Sampling from Ames, Iowa

If you have access to data on an entire population, say the size of every house in Ames, Iowa, it's straight forward to answer questions like, "How big is the typical house in Ames?" and "How much variation is there in sizes of houses?". If you have access to only a sample of the population, as is often the case, the task becomes more complicated. What is your best guess for the typical size if you only know the sizes of several dozen houses? This sort of situation requires that you use your sample to make inference on what your population looks like.

The data

In the previous lab we looked at the population data of houses from Ames, Iowa. Let's start by loading that data set.

```
load(url("http://www.openintro.org/stat/data/ames.RData"))
```

In this lab we'll start with a simple random sample of size 60 from the population. Specifically, this is a simple random sample of size 60. Note that the data set has information on many housing variables, but for the first portion of the lab we'll focus on the size of the house, represented by the variable `Gr.Liv.Area`.

```
population = ames$Gr.Liv.Area  
samp = sample(population, 60)
```

Exercise 1 Make a histogram of 'samp' and describe the distribution of the histogram in detail. To change the number of stripes in the histogram, you can use the `breaks` code you learned earlier.

Exercise 2 Describe the distribution of your sample. What would you say is the "typical" size within your sample? Also state precisely what you interpreted "typical" to mean.

Exercise 3 Would you expect another student's distribution to be identical to yours? Would you expect it to be similar? Why or why not?

Confidence intervals

One of the most common ways to describe the typical or central value of a distribution is to use the mean. In this case we can calculate the mean of the sample using,

```
sample_mean = mean(samp)
```

Return for a moment to the question that first motivated this lab: based on this sample, what can we infer about the population? Based only on this single sample, the best estimate of the average living area of houses sold in Ames would be the sample mean, usually denoted as $\bar{x}$ (here we're calling it `sample_mean`). That serves as a good *point estimate* but it would be useful to also communicate how uncertain we are of that estimate. This can be captured by using a *confidence interval*.

We can calculate a 95% confidence interval for a sample mean by adding and subtracting 1.96 standard errors to the point estimate.

Exercise 4 Based on your sample, write code to calculate and display your 95% confidence interval for the true average size of houses in Ames (population mean size of houses in Ames). Hint: compute standard deviation first.

Confidence levels

Exercise 5 What does “95% confidence” mean?

In this case we have the luxury of knowing the true population mean since we have data on the entire population. This value can be calculated using the following command:

```
mean(population)
```

Exercise 6 Does your confidence interval capture the true average size of houses in Ames? Get a new sample, does the confidence interval of the mean of your new sample capture this value?

Exercise 7 If you obtain many samples and compute their corresponding 95% confidence intervals for the population mean, what proportion of those intervals would you expect to capture the true population mean? Why?

Using R, we're going to recreate many samples to learn more about how sample means and confidence intervals vary from one sample to another. *Loops* come in handy here .

Here is the rough outline:

- Obtain a random sample.
- Calculate the sample's mean and standard deviation.
- Use these statistics to calculate a confidence interval.
- Repeat steps (1)-(3) 50 times.

But before we do all of this, we need to first create empty vectors where we can save the means and standard deviations that will be calculated from each sample. And while we're at it, let's also store the desired sample size as `n` .

```
samp_mean = rep(NA, 50)
samp_sd = rep(NA, 50)
n = 60
```

Now we're ready for the loop where we calculate the means and standard deviations of 50 random samples.

```
for(i in 1:50){
  samp = sample(population, n) # obtain a sample of size n = 60 from the population
  samp_mean[i] = mean(samp)    # save sample mean in ith element of samp_mean
  samp_sd[i] = sd(samp)        # save sample sd in ith element of samp_sd
}
```

Lastly, we construct the confidence intervals.

```
lower_vector = samp_mean - 1.96 * samp_sd / sqrt(n)
upper_vector = samp_mean + 1.96 * samp_sd / sqrt(n)
```

Lower bounds of these 50 confidence intervals are stored in `lower_vector`, and the upper bounds are in `upper_vector`. Let's view the first interval.

```
c(lower_vector[1], upper_vector[1])
```

Additional Exercises

Exercise 8 Obtain 100 new samples, calculate and plot their 95% confidence intervals following the instructions above. What proportion of your intervals capture the population mean? Is this proportion exactly equal to the confidence level? If not, explain why.

```
plot_ci(lower_vector, upper_vector, mean(population))
```

Exercise 9 Pick a confidence level of your choosing, provided it is not 95%. What is the appropriate critical value?

Exercise 10 Calculate 50 confidence intervals at the confidence level you chose in the previous question. You do not need to obtain new samples, simply calculate new intervals based on the sample means and standard deviations you have already collected. Using the `plot_ci` function, plot all intervals and calculate the proportion of intervals that include the true population mean. How does this percentage compare to the confidence level level selected for the intervals?

Submission

When you are done, regarding the submission of the lab report, please note the following:

1. The report needs to include Exercise1-10;

2. Some exercises may not require coding;
3. Please enter your student IDs and names of your group at the top of the report;
4. Please submit the lab report within 24 hours after class;
5. The submitted code must be able to run successfully at one time;
6. Please complete the lab report **in HTML and RMD formats, package the file into a .zip compressed file and upload it to the Blackboard learning (BB) system**, and for each group only one lab report is needed;
7. Please name your file following the format: "Student ID 1-Name 1-Student ID 2-Name 2_Lab Number_Course Name". For example, 2021XXXXXXXX- 张三 -2021XXXXXXXX- 李四 _Lab5_Data-Analysis-and-Modeling..

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